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should indicates a recommendation to do something

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can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the digital test sequences for the Immersive Voice and Audio Services (IVAS) codec. These sequences shall be used in conformance testing for implementations of the IVAS codec (3GPP TS 26.253), Rendering (3GPP TS 26.254), Error Concealment of Lost Packets (3GPP TS 26.255) and Jitter Buffer Management (JBM) (3GPP TS 26.256) and its reference C code specification, which are 3GPP TS 26.258 (floating-point) and 3GPP TS 26.251 (fixed-point). The sequences shall also be used for conformance testing of implementations of Immersive Audio for Split Rendering Scenarios (ISAR) according to 3GPP TS 26.249. In addition, the present document specifies procedures for conformance testing.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
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- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 26.250: "Codec for Immersive Voice and Audio Services (IVAS); General overview".

[3] 3GPP TS 26.251: "Codec for Immersive Voice and Audio Services (IVAS); C code (fixed point)".

[4] 3GPP TS 26.253: "Codec for Immersive Voice and Audio Services (IVAS); Detailed Algorithmic Description including RTP payload format and SDP parameter definitions".

[5] 3GPP TS 26.254: "Codec for Immersive Voice and Audio Services (IVAS); Rendering".

[6] 3GPP TS 26.255: "Codec for Immersive Voice and Audio Services (IVAS); Error concealment of lost packets".

[7] 3GPP TS 26.256: "Codec for Immersive Voice and Audio Services (IVAS); Jitter Buffer Management".

[8] 3GPP TS 26.258: "Codec for Immersive Voice and Audio Services (IVAS); C code (floating-point)".

[9] 3GPP TS 26.444: "Codec for Enhanced Voice Services (EVS); Test Sequences".

[10] ETSI TS 103 634 V1.6.1 (2025-10), "Digital Enhanced Cordless Telecommunications (DECT); Low Complexity Communication Codec plus (LC3plus)".

[11] 3GPP TS 26.249: "Immersive Audio for Split Rendering Scenarios (ISAR)".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

AMR-WB	Adaptive Multi Rate Wideband (codec)
EVS	Enhanced Voice Services (codec)
IVAS	Immersive Voice and Audio Services (codec)
ISAR	Immersive Audio for Split Rendering Scenarios
JBM	Jitter Buffer Management

4 General

4.1 Introduction

This specification provides digital test sequences that shall be used to test conformance for an implementation of the IVAS codec (TS 26.253 [4]), Rendering (TS 26.254 [5]), Error Concealment of Lost Packets (TS 26.255 [6]) and Jitter Buffer Management (JBM) (TS 26.256 [7]), and its reference C code specification, which are TS 26.258 [8] (floating-point) and TS 26.251 [3] (fixed-point). An overview of the IVAS Codec specifications is found in TS 25.250 [2]. The sequences shall also be used for conformance testing of implementations of Split Rendering functions addressing Immersive Audio for Split Rendering Scenarios (ISAR) according to TS 26.249 [11].

A standard compliant implementation of the above specifications shall pass the conformance tests according to clause 7. The necessary test sequences can be found in the corresponding ZIP files according to the attached Readme.txt file.

NOTE: The test sequences apply to specific version(s) of the IVAS codec as indicated by the name of the ZIP file, e.g., IVAS-FL-1.0. The codec version is used to have consistent numbering across reference C code specifications.

Clause 5 describes the format of the files, which contain the digital test sequences. Clause 6 describes the test sequences for the IVAS codec, including rendering, error concealment of lost packets, jitter buffer management and split rendering. Clause 7 describes the conformance testing procedure for implementations of the IVAS codec.

5 Test sequence format

5.1 Introduction to test sequence format

This clause provides information on the format of the digital test sequences for the IVAS codec (TS 26.253 [4]), Rendering (TS 26.254 [5]), Error Concealment of Lost Packets (TS 26.255 [6]) and Jitter Buffer Management (JBM) (TS 26.256 [7]) and its reference C code specifications, which are TS 26.258 [8] (floating-point) and TS 26.251 [3] (fixed-point).

5.2 File format

The test sequence data is provided in PC (little-endian byte order) files, according to table 1.

Table 1: Overview of test sequence files

File type	File extensions
Audio input to the encoder and output from the decoder and renderer	*.wav
ISM metadata, Head rotation trajectories	*.csv
MASA metadata	*.met
Rate switching	*.bin
IVAS bitstreams	*.192
Network simulator streams	*.netsimout
ISAR bitstreams	*.bit
ISAR bitstreams with frame errors	*.ep
IVAS bitstreams with frame errors	*.fer
Renderer configuration (text format or binary format)	*.cfg, *.dat
Renderer scene description	*.txt

6 IVAS codec test sequences including error concealment of lost packets

6.1 Introduction to test sequences

This clause provides information on the test sequences designed to exercise the IVAS codec.

6.2 Codec configuration

The codec shall be configured according to the instructions in Readme_IVAS_{enc, dec, rend, JBM_dec, ISAR_dec, ISAR_post_rend}.txt for each test case respectively in accordance with clause 6.3. For the bit-exact EVS compatibility mode of IVAS, including the AMR-WB interoperable function, the codec shall be configured in accordance with TS 26.444 [9].

6.3 IVAS codec test sequences

6.3.1 Mono operation test sequences

For mono operation (utilizing the bit-exact EVS compatibility mode of IVAS, including the AMR-WB interoperable function) the encoder and decoder shall be tested using test sequences and instructions in accordance with TS 26.444 [9].

6.3.2 Encoder test sequences

To test an IVAS encoder (beyond mono operation, see clause 6.3.1), test sequences and instructions provided in Readme_IVAS_enc.txt shall be used.

6.3.3 Decoder test sequences

To test an IVAS decoder (beyond mono operation, see clause 6.3.1), test sequences and instructions provided in Readme_IVAS_dec.txt shall be used. To test the IVAS decoder for split rendering (ISAR pre-renderer), test sequences and instructions provided in Readme_IVAS_ISAR_dec.txt shall be used.

6.3.4 Renderer test sequences

To test an IVAS renderer, test sequences and instructions provided in Readme_IVAS_rend.txt shall be used.

6.3.5 Jitter buffer management test sequences

To test jitter buffer management (JBM) for an IVAS decoder, test sequences and instructions provided in Readme_IVAS_JBM_dec.txt shall be used.

6.3.6 Split rendering post renderer test sequences

To test post renderer for IVAS split rendering, test sequences and instructions provided in Readme_IVAS_ISAR_post_rend.txt shall be used.

7 Conformance Testing

7.1 Bit-exact Conformance

For an implementation to be declared conformant according to the bit-exact conformance test procedure, the output sequences of the corresponding feature being implemented (IVAS encoder, IVAS decoder, IVAS renderer, JBM, ISAR pre-renderer, ISAR post-renderer) shall match bit-exactly the reference test sequences provided in the corresponding ZIP files in accordance with clause 6, including clause 6.3.1 for mono operation of the IVAS encoder and IVAS decoder. This applies for all implementations of the IVAS codec (TS 26.253 [4]), Rendering (TS 26.254 [5]), Error Concealment of Lost Packets (TS 26.255 [6]) and Jitter Buffer Management (JBM) (TS 26.256 [7]), and its reference C code specifications, which are TS 26.258 (floating-point) and TS 26.251 (fixed-point). This also applies for implementations of pre-rendering and post-rendering functions addressing Immersive Audio for Split Rendering Scenarios (ISAR) according to TS 26.249 [11].

If optional features are implemented, the corresponding conformance tests shall pass with bit-exact results.

7.2 Non-bit-exact Conformance

7.2.1 Non-bit-exact Conformance for IVAS floating-point operations

7.2.1.1 IVAS floating-point mono operation

For IVAS mono operation (of and IVAS encoder or IVAS decoder), if an implementation under test is based on floating–point code (TS 26.258 [8]) and the output sequences are not bit-exact to the test sequences according to clause 6, the non-bit-exact conformance testing procedure defined in TS 26.444 [9] shall be used to test the conformance.

If optional features are implemented, the corresponding conformance tests shall pass with bit-exact results.

7.2.1.2 IVAS floating-point conformance in stereo and immersive operation

If an implementation under test is based on the reference floating–point code (TS 26.258 [8]) and the output sequences are not bit-exact to the test sequences according to clause 6, the non-bit-exact conformance testing process defined here shall be used to test the conformance.

A conformant floating-point implementation of the IVAS codec shall be compliant to the reference specification in TS 26.250 [3] by implementing all the algorithmic steps of the IVAS codec, further specified in 3GPP TS 26.253 (Detailed Algorithmic Description) [4], 3GPP TS 26.254 (IVAS rendering) [5], 3GPP TS 26.255 (Packet Loss Concealment (PLC) of Lost Packets) [6], 3GPP TS 26.256 (Jitter Buffer Management (JBM)) [7] and 3GPP TS 26.249 Immersive Audio for Split Rendering Scenarios (ISAR) [11].

If a floating-point implementation uses the Jitter Buffer Management (JBM) according to TS 26.256 [7], the implementation shall be compliant to the reference specification in TS 26.253 [4] by implementing all the algorithmic steps of 3GPP TS 26.256 (Jitter Buffer Management (JBM)).

If a floating-point implementation uses the renderer according to TS 26.254 [5], the implementation shall be compliant to the reference specification in TS 26.253 [4] by implementing all the algorithmic steps of TS 26.254 (IVAS rendering).

If a floating-point implementation uses pre-rendering and post-rendering functions addressing Immersive Audio for Split Rendering Scenarios (ISAR) according to TS 26.249 [11], the implementation shall be compliant to the reference specification in TS 26.253 [4] by implementing all the algorithmic steps of TS 26.249 (Immersive Audio for Split Rendering Scenarios (ISAR)).

An implementation shall be tested for non-bit-exact conformance using two specific tests:

- Decoder test comparing the implementation decoder with TS 26.258 [8] decoder.
- Encoder test comparing the implementation encoder with TS 26.258 [8] encoder.

Both encoder and decoder tests shall pass for the implementation to be declared conformant. Figure 7.1 shows the flow chart of the non-bit-exact conformance process.

In addition to the encoder and decoder conformance tests, if an implementation uses the Jitter Buffer Management (JBM) according to TS 26.256 [7], then it shall be tested for non-bit-exact conformance using:

- Jitter Buffer Management (JBM) test comparing the implementation JBM with TS 26.258 [8] decoder.

In addition to the encoder and decoder conformance tests, if an implementation uses the IVAS rendering according to TS 26.254 [5], then it shall be tested for non-bit-exact conformance using:

- Renderer test comparing the implementation encoder with TS 26.258 [8] renderer.

In addition to the encoder and decoder conformance tests, if an implementation uses pre-rendering and post-rendering functions addressing Immersive Audio for Split Rendering Scenarios (ISAR) according to TS 26.254 [5], then it shall be tested for non-bit-exact conformance using the following specific tests:

- Pre-renderer test comparing the implementation pre-renderer with TS 26.258 [8] decoder and renderer
- Post-renderer test comparing the implementation post-renderer with TS 26.258 [8] post-renderer.

Non-bit-exact conformance is defined based on comparisons of WAV output files and metadata output files. Metadata output files are ISM metadata (with .csv file extension) and MASA metadata (with .met file extension) output files.

The tests with WAV output files are described in detail in Annex A.

The tests with Metadata output files shall yield a maximum deviation from the reference that is not larger than the threshold specified in Annex A.3.1.2, where the maximum deviation is measured separately per metadata parameter for each metadata format.

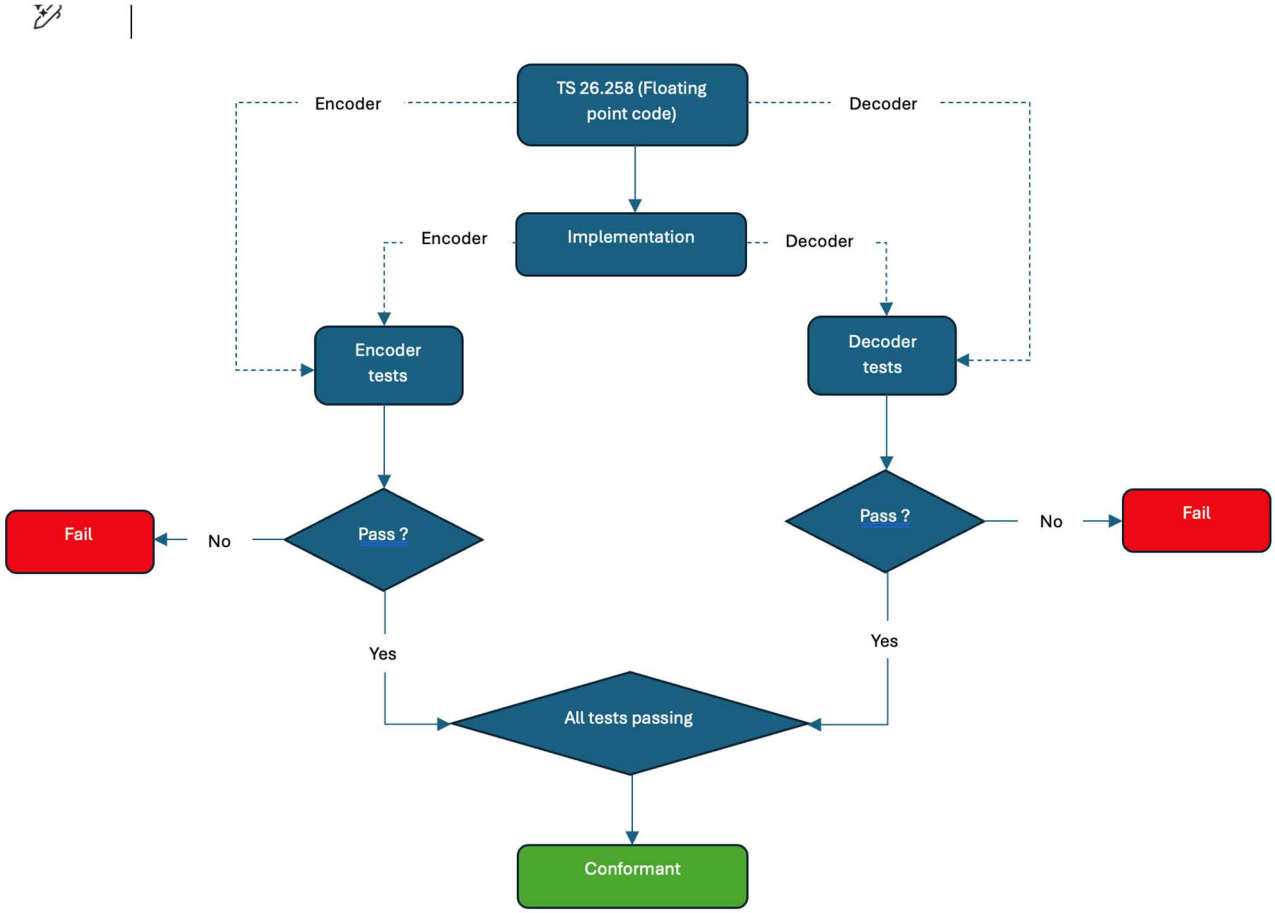


Figure 7.1: Non-bit-exact conformance test

7.2.3 Running the Tests

The ZIP package containing floating-point test sequences includes scripts and executable utilities for running bit-exact and non-bit-exact conformance tests. Comprehensive instructions are provided within the package outlining the process for running these tests. Additional information on running the tests can be found in Annex A. In the case of discrepancy between the procedures described in Annex A and those implemented by the scripts in the ZIP file, the procedures executed by within the scripts prevail.

For all IVAS tests, including encoder, decoder, JBM, renderer and ISAR pre- and post-renderer, instructions for the implementation under test (CUT) are available in the README.md text file. The scripts require specific Python® packages (installation instructions provided in the README.md file) as well as additional tools, such as `wav-diff` and `masaDiffTool`. To execute IVAS conformance test, run the `runConformance.py` Python® script.

The ZIP file includes MLD values (corridor) for all reference test vectors and reference metadata files in the `testvec\testv\ref_mld` folder. An implementation under test (CUT) is conformant upon successful completion of all encoder and decoder tests. Additionally, an implementation under test (CUT) is conformant for JBM, Rendering and Split-rendering (ISAR) by successfully passing their respective test suites.

7.3 LC3plus Conformance

For IVAS/ISAR split rendering operation utilizing LC3plus, the LC3plus encoder and decoder implementation shall pass the conformance test according to ETSI TS 103 634 clause 7. The conformance tests configuration is defined in Table 2.

Table 2: Conformance tests for LC3plus in IVAS split rendering

LC3plus for IVAS split rendering	Codec configuration				Conformance tests group (see clause 7.3.1) (see note 1)		
	HR mode	**Sampling rate [Hz]**	**Frame size [ms]**	**Bit rate [bytes per frame] (see note 2)**	**Core coder**	**Concealment**	**Channel Coder**
	Disabled	48 000	5	80, 120, 160	Enc, Dec, EncDec	Dec	N/A
			10	155[3], 240, 320, 400			
	Enabled		5	80, 120, 160	Enc, Dec, EncDec	Dec	N/A
			10	155[3], 240, 320, 400			
NOTE 1: Some conformance tests are conducted for the modules encoder (enc), decoder (dec) and codec (EncDec) separately. NOTE 2: As LC3plus operates in dual-mono for stereo signals, the conformance tests are applied on mono signals only. NOTE 3: Rates correspond to Basic Audio Profile for Bluetooth Low Energy audio using the configurations 48_6							

Annex A: Tools Description (normative)

A.1 Decoder Test

A.1.1 General Considerations

The WAV outputs and associated MASA format and ISM format metadata outputs from the decoder under test (CUT) are obtained by running the reference bit-streams included in the ZIP package through the testing system described in Figure A.1, following the testing procedure described in clause 7.2.1.2. The reference decoder is built from the floating-point code of TS 26.258 [8].

NOTE: The reference decoder must be built following the procedures and for the platforms as specified in TS 26.258 [8] or TS 26.250 [3], respectively.

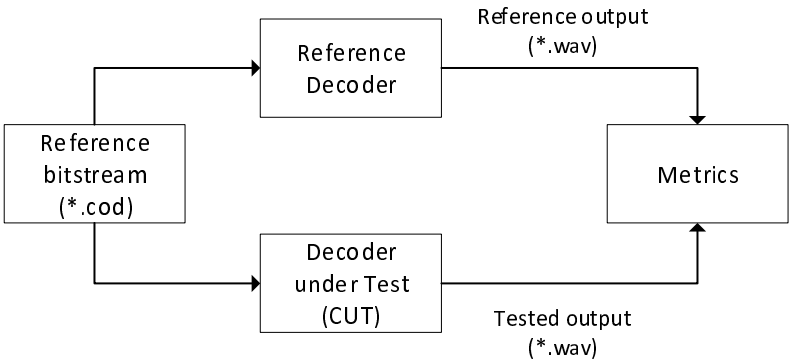


Figure A.1: Flow diagram for the decoder test using signal-based metrics

The MLD metric, as defined in clause A.1.2, is used to test the non-bit-exact decoder floating-point conformance. The MLD metric is calculated on the reference WAV output signal $x_{REF}(t)$ and the tested WAV output signal $x_{TST}(t)$ based on 20ms frames. The frames of the two signals will be time aligned; this means the delay compensation in IVAS encoder and decoder remains ON (the default configuration).

The number of samples N for a 20ms frame size is defined by $N = f_s \cdot 0.02$, where f_s represents the sampling rate.

Metadata parameters are tested using per-frame metadata parameter differences as described in A.1.2. The metrics are calculated between reference metadata $m_{REF}(t)$ and $m_{TST}(t)$ on 20ms frame steps. Metadata will follow the alignment of the corresponding WAV signal.

A.1.2 Metrics

The Maximum Loudness Difference (MLD) is used as defined in A.2.2 of [9]. Given that the IVAS decoder output can have multiple channels, the MLD per 20 ms frame is computed as the maximum of MLD values of all the channels in that frame.

To achieve conformance, the per-frame MLD values of all tested sequences of the implementation under test (CUT) shall be lower than a threshold $Th_{ref}[n]$ defined as

$$Th_{ref}[n] = 0.1 + MLD_{ref}[n]$$

where $MLD_{ref}[n]$ is the reference per-frame MLD value and 0.1 represents a fixed headroom. Thus, for each tested output the MLD is defined as

$$MLD = min\left(Th_{ref}[n] - N_{diff}[n]\right)$$

where, $N_{diff}[n]$ is the calculated per-frame MLD value as defined in A.2.2 of [9].

The test output will be considered equivalent to the reference output if the MLD value is positive or equal to 0, i.e. the Maximum Loudness Difference does not exceed the threshold for any of the frames.

The $MLD_{ref}[n]$ values are defined by running reference implementations of the IVAS floating-point C code on various reference platforms listed in clause A.4.

Thus, for each test sequence, the $MLD_{ref}[n]$ values of all reference implementations define a so-called 'corridor', or 'refline'. This then leads to a profile for each test sequence, which contains on a 20ms frame basis an allowed MLD value relative to its reference. Allowed differences in implementations under test (CUTs) are thus limited to the tolerable differences by the different compilers used for the corridor generation.

MASA metadata frames are tested for descriptive and spatial metadata equivalence. For all metadata frames, the descriptive metadata of the m_{REF} and m_{TST} need to be identical for the test to pass. For spatial metadata, a maximum and mean parameter difference values are formed per 20 ms frame for each of the spatial metadata parameters defined in TS 26.258 [8] Annex A: direction index, direct-to-total ratio, diffuse-to-total ratio, remainder-to-total ratio, spread coherence, and surround coherence.

Before calculating difference, the direction index is decoded into azimuth and elevation pair (ϕ, θ) as detailed in TS.26.258 [8] Annex A.5. Then each $(\phi, \theta)_{REF}$ and $(\phi, \theta)_{TST}$ are compared in spatial direction by converting the azimuth and elevation pairs into cartesian coordinates with

$$\boldsymbol{v} = \begin{pmatrix} v_x = \cos(\theta)\cos(\phi) \\ v_y = \cos(\theta)\sin(\phi) \\ v_z = \sin(\theta) \end{pmatrix}$$

and then calculating spatial difference angle with

$$D_{dir} = |\mathrm{acos}(\max(-1.0, \min(1.0, \boldsymbol{v}_{REF} \cdot \boldsymbol{v}_{TST})))|$$

The above difference is computed for each direction parameter set present (1 or 2) and each subframe and frequency band. The value D_{dir} is further scaled into numeric range of [0,1] by dividing the acos-output by π. This allows similar difference range as with the ratio parameters.

For all other spatial parameters, the following ratio difference is calculated

$$D_{ratio} = |r_{REF} - r_{TST}|$$

For direct-to-total ratio and spread coherence, this is calculated for each direction parameter set (1 or 2) and each subframe and frequency band. For diffuse-to-total ratio, remainder-to-total ratio, and surround coherence this is calculated for each subframe and frequency band.

For each frame and parameter difference D, the maximum difference and mean difference are calculated with

$$D_{max} = \max_{\substack{d \in (1,2), \\ s \in (1,2,3,4), \\ b \in (1,2,\ldots,24)}} \big(D(d,s,b)\big)$$

$$D_{mean} = \frac{1}{192} \sum_{d=1}^{2} \sum_{s=1}^{4} \sum_{b=1}^{24} D(d,s,b)$$

for parameters with two direction sets present and with

$$D_{max} = \max_{\substack{s \in (1,2,3,4), \\ b \in (1,2,\ldots,24)}} \big(D(s,b)\big)$$

$$D_{mean} = \frac{1}{96} \sum_{s=1}^{4} \sum_{b=1}^{24} D(s,b)$$

for common spatial metadata parameters and parameters with one direction set present. Here d is the direction set index, s is the time subframe, and b is the frequency band.

After forming these per-frame metadata difference metrics, the metadata difference metrics are compared against a threshold, that is,

$$Th_{MASA}(n) = 0.004 + D_{ref}(n)$$

and for each file and frame, any per-frame metadata difference metric $D_{max}(n)$ or $D_{mean}(n)$ needs to be smaller than or equal to the threshold. This threshold can be defined for each metadata parameter separately or a common threshold is used.

$D_{ref}(n)$ values are defined by running reference implementations of the IVAS floating-point C code on various reference platforms listed in clause A.4.

Thus, for each frame of each test sequence, the maximum of per-frame metadata difference metric values of all reference implementations define a so-called 'corridor', or 'refline'. This then leads to a profile for each IVAS test sequence, which contains on a 20ms frame basis an allowed $D_{ref}(n)$ value relative to its reference. Allowed differences in implementations under test (CUTs) are thus limited to the tolerable differences by the different compilers used for the corridor generation.

All test sequences need to pass both the MLD test and the metadata test for the decoder implementation under test (CUT) to be conformant.

A.2 Encoder Test

A.2.1 General Consideration

The WAV outputs from the encoder under test (CUT) are obtained by running the reference input signals included in the ZIP package through the testing system described in Figure A.3 following the testing procedure described in clause 7.2.1.2.

NOTE: The reference encoder and the reference decoder must be built following the procedures and for the platforms as specified in TS 26.258 [8] or TS 26.250 [3], respectively.

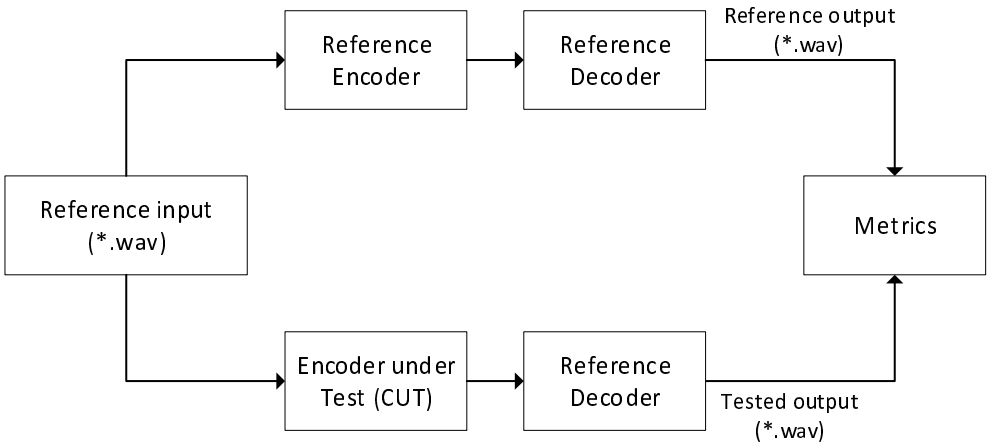


Figure A.3: Flow diagram for the encoder test using MLD Loudness Difference metric.

The MLD metric, as defined in clause A.1.2, is used to test the non-bit-exact encoder floating-point conformance. The MLD metric is calculated on the reference WAV output signal $x_{REF}(t)$ and the tested WAV output signal $x_{TST}(t)$ based on 20ms frames. The frames of the two signals will be time aligned; this means the delay compensation in IVAS encoder and decoder remains ON (the default configuration).

The number of samples N for a 20ms frame size is defined by $N = f_s \cdot 0.02$, where f_s represents the sampling rate.

A.2.2 Metrics

As defined in A.1.2.

All test sequences need to pass both the MLD test and the metadata test for the encoder implementation under test (CUT) to be conformant.

A.3 Thresholds and Criteria

A.3.1 Thresholds and Criteria for IVAS floating-point operations

A.3.1.1 MLD Thresholds for WAV comparisons

The per-frame threshold $Th_{ref}[n]$ is used to define a permissible corridor of MLD values for the implementation under test (CUT) as defined in clause A.1.2.

A.3.1.2 Thresholds for Metadata comparisons

For Metadata output files of IVAS floating-point operations a deviation threshold is defined for each metadata format separately.

For ISM metadata format, each metadata parameter is defined to have a separate threshold. The threshold of 0 applies to each metadata parameter.

For MASA metadata format, the metadata parameter difference metric corridors defined in A.1.2 are used to define permissible per-frame threshold $Th_{MASA}(n)$. For each metadata parameter, additional tolerance of 0.004 (approximately 1/255 quantization step) is included in the threshold.

A.4 Reference Implementations

To calculate the MLD corridor, which defines the threshold values for non-bit-exact conformance tests, a set of reference implementations was used. Table A.8 lists these implementations, including compiler, target platform and compiler settings. These implementations are based on mainstream compilers and platforms and use the reference release of the IVAS floating-point source code version specified in TS 26.258 [8]. They are not bit-exact with each other nor with the reference implementation (Ubuntu™ 24.04, Clang 18, O0).

Table A.4: List of Reference Implementations

Name	Processor	Compiler	Optimization	OS
ubuntu_gcc_v13_o0	Intel® Core™ i7-12800H	GCC v13.3.0	O0	Ubuntu™ 24.04
ubuntu_gcc_v13_o2	Intel® Core™ i7-12800H	GCC v13.3.0	O2	Ubuntu™ 24.04
ubuntu_gcc_v13_o3	Intel® Core™ i7-12800H	GCC v13.3.0	O3	Ubuntu™ 24.04
ubuntu_clang_v18_o2	Intel® Core™ i7-12800H	Clang v18.1.3	O2	Ubuntu™ 24.04
ubuntu_clang_v18_o3	Intel® Core™ i7-12800H	Clang v18.1.3	O3	Ubuntu™ 24.04
ubuntu_arm64_clang_v18_o0	ARM® 64-bit	Clang v18.1.3	O0	Ubuntu™ 24.04
ubuntu_arm64_clang_v18_o2	ARM® 64-bit	Clang v18.1.3	O2	Ubuntu™ 24.04
ubuntu_arm64_clang_v18_o3	ARM® 64-bit	Clang v18.1.3	O3	Ubuntu™ 24.04
msvc2017_v141_Debug_O0	Intel® Xeon® E5-2680	MSVC 2017 Toolset: v141 Compiler: 19.16.27051 Target platform: Windows® 10.0.17763.0	O0	Windows® Server 2019
msvc2017_v141_Release_O2	Intel® Xeon® E5-2680	MSVC 2017 Toolset: v141 Compiler: 19.16.27051 Target platform: Windows® 10.0.17763.0	O2	Windows® Server 2019
msvc_19.41_arm_O0	ARM® 64-bit	MSVC 2022 Toolset: v143 Compiler: 19.41.34120.0 Target platform: Windows® 10.0.26200	O0	Windows® 10
msvc_19.41_arm_O2	ARM® 64-bit	MSVC 2022 Toolset: v143 Compiler: 19.41.34120.0 Target platform: Windows® 10.0.26200	O2	Windows® 10
msvc_19.41_arm_O3	ARM® 64-bit	MSVC 2022 Toolset: v143 Compiler: 19.41.34120.0 Target platform: Windows® 10.0.26200	O3	Windows® 10
mac_arm_clang_v16_o0	ARM® Cortex®-M4	Xcode® 16.2, Apple® Clang 16.0.0.16000026	O0	MacOS® 15.7.4
mac_arm_clang_v16_o2	ARM® Cortex®-M4	Xcode® 16.2, Apple® Clang 16.0.0.16000026	O2	MacOS® 15.7.4
mac_arm_clang_v16_o3	ARM® Cortex®-M4	Xcode® 16.2, Apple® Clang 16.0.0.16000026	O3	MacOS® 15.7.4

Annex B (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2023-11	SA4#126	S4-231880				Initial version	0.0.1
2023-11	SA4#126	S4-231994				Presented to SA4 plenary	0.1.0
2023-12	SA#102	SP-231302				Version 1.0.0 created by MCC	1.0.0
2024-02	SA4#127	S4-240343				Added initial test sequences for TS 26.258	1.1.0
2024-03	SA#103	SP-240028				Version 2.0.0 created by MCC	2.0.0
2024-03						Version 18.0.0 created by MCC	18.0.0
2024-06	SA#104	SP-240693	0001	2	B	Updated IVAS test sequences	18.1.0
2024-06						Title changed, abbreviation "IVAS" added per TSG SA decision	18.1.0
2024-06	SA#105	SP-241115	0002	1	F	Correction of references	18.2.0
2025-10	-	-	-	-	-	Update to Rel-19 version (MCC)	**19.0.0**
2025-12	SA#110	SP-251431	0004	2	A	Corrections for IVAS test sequences	**19.1.0**
2025-12	SA#110	SP-252025	0005	2	B	Updates related to IVAS conformance testing of TS 26.258 and TS 26.251	**19.1.0**
2026-06	SA#112	SP-260393	0007	1	F	Corrections for IVAS test sequences	**19.2.0**

History

Version	Date	Status
V19.0.0	November 2025	Publication
V19.1.0	February 2026	Publication
V19.2.0	August 2026	Publication